# Project Name: Debounced Search with Auto-Suggestions

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### Overview

This project involves implementing a search bar with auto-suggestions using React. The search bar fetches recipe data dynamically from an API and displays real-time suggestions as the user types. To optimize performance, the implementation includes debouncing to reduce unnecessary API calls.

### Features

**1. Fetching Recipe Data from API**

- Retrieves search results from https://dummyjson.com/recipes/search?q={query}.
- Uses useEffect to make API calls based on user input.
- Stores fetched recipes in state for rendering suggestions.

**2. Search Input with Real-Time Suggestions**

- Users can type into a search field to find recipes.
- Real-time recipe suggestions appear in a dropdown list.

**3. Debouncing for Optimized API Calls**

- Implements a 500ms delay before triggering API requests.
- Uses setTimeout for debouncing.
- Clears the previous timer using clearTimeout to prevent redundant calls.

**4. Clickable Suggestions for Selection**

- Users can click on a suggested recipe to select it.
- The selected recipe’s details are displayed below the search bar.

**5. No Results Handling**

- If no matching recipes are found, displays "No recipes found."
- Ensures case-insensitive search behavior.

### Technologies Used

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| **React.js** | Frontend development framework for building UI components. |
| **JavaScript** | Used for logic implementation, async operations, and event handling. |

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| **React Hooks (useState, useEffect)** | Manages state and side effects efficiently. |
| **Fetch API** | Handles API communication for retrieving recipe data. |
| **CSS (Styled Components / Tailwind CSS)** | Enhances the UI and improves user experience. |
| **setTimeout & clearTimeout** | Implements debouncing to optimize API calls. |

**Implementation Details**

**1. State Management**

- **Uses useState for:**
  - **searchQuery (to store user input).**
  - **suggestions (to store fetched recipe suggestions).**
  - **selectedRecipe (to store details of the selected recipe).**

**2. API Fetching & Debouncing**

- **useEffect listens for changes in searchQuery.**
- **Uses setTimeout inside useEffect to delay API requests by 500ms.**
- **Calls clearTimeout before setting a new timer to optimize performance.**
- **Fetches data from https://dummyjson.com/recipes/search?q={query} dynamically.**

**3. User Interactions**

- **onChange event updates searchQuery in real-time.**
- **onClick event allows users to select a suggestion.**
- **onKeyPress ensures smooth user experience while typing.**
- **Hover effects implemented using CSS for better UX.**

**4. Component Structure**

- **SearchBar Component:**
  - **Renders an input field and dropdown for suggestions.**
  - **Handles user input and triggers API calls with debouncing.**
- **SuggestionsList Component:**
  - **Displays real-time suggestions fetched from the API.**
  - **Handles click events to select a recipe.**
- **RecipeDetails Component:**
  - **Displays selected recipe details upon user selection.**

**5. Performance Optimization**

- **Uses memoization techniques to avoid unnecessary re-renders.**
- **Implements efficient state management to reduce computation overhead.**
- **Ensures API calls are minimized using debouncing to improve application efficiency.**

**Conclusion**

The Debounced Search with Auto-Suggestions project enhances user experience by efficiently fetching and displaying search results. By integrating debouncing, it minimizes redundant API requests while maintaining real-time interactivity. The use of React hooks, API optimization, and efficient event handling ensures a seamless search experience.